Working paper | Life and health research

Mortality of the Quantified Self
A Bayesian Credibility Framework for Wearable-Derived Life Underwriting

A coupled biometric–statistical–regulatory framework joining continuous wearable telemetry, classical credibility theory, the Holmström informativeness principle, and the privacy-regime constraints of POPIA, GDPR, and HIPAA.

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Abstract

Continuous biometric telemetry from consumer wearables — Apple Watch, Fitbit, Garmin, Oura, Whoop — now generates sub-second resolution time series of resting heart rate, heart-rate variability (HRV), sleep architecture, blood oxygen saturation (SpO_2), respiration rate, and $\dot{V}O_2$-max proxies for an installed base exceeding 300 million users globally. Major life insurers (John Hancock Vitality, Manulife Move, Discovery Vitality, Prudential PRUActive) have begun integrating these signals into underwriting and renewal pricing. The actuarial question is no longer whether to use wearable telemetry but *at what credibility weight, under what privacy regime, and with what fairness constraints*.

We propose a coupled biometric–statistical–regulatory framework for wearable-derived life underwriting with the following structure. (i) A **multi-signal informativeness audit** that ranks resting HR, nightly HRV, sleep efficiency, SpO_2 nadir, and step-count distribution by their incremental information content with respect to the all-cause mortality intensity, following the Holmström informativeness principle. (ii) A **Bayesian credibility formulation** in which the wearable-derived prior intensity $\lambda_0(\mathbf{w})$ is updated continuously with claims experience to yield a posterior intensity $\lambda(t)$; we derive the exact credibility weight $Z(t)$ and its asymptotic behaviour. (iii) An **exponential-tilt premium adjustment** that maps the posterior intensity into a multiplicative life-premium factor with closed-form sensitivity. (iv) A **regulatory comparison** of the admissibility, consent, and rating-band constraints under POPIA (South Africa), GDPR (EU), HIPAA (USA), and the New York DFS Circular Letter 2019-1 on accelerated underwriting. (v) A worked example pricing a 25-year-old applicant with twelve months of continuous Apple Watch telemetry against the standard underwriting class.

The framework explicitly addresses three failure modes that current industry implementations ignore: **selection-on-wearable ownership** (wearable owners are a non-representative slice of the insured population), **signal substitution bias** (an applicant can game the contract by wearing the device only when rested), and **intergenerational fairness** (the wearable underwriting regime should not implicitly penalise the next generation of policyholders who have spent more of their life under surveillance).

Keywords wearable telemetry; Bayesian credibility; Holmström informativeness; life underwriting; accelerated underwriting; POPIA; GDPR; selection bias; actuarial fairness; Vitality.

JEL / MSC G22, D82, K24, 62P05, 91B30.

Contents

1. Introduction

1.1. The wearable underwriting moment

By the end of 2025, the installed base of consumer wearables capable of continuous biometric telemetry had exceeded 300 million devices. Apple Watch, Fitbit (Google), Garmin, Oura, and Whoop together account for the majority of this volume, with rapid market share gains by ring-form-factor devices (Oura, Ultrahuman, RingConn) that prioritise nightly metrics over fitness tracking. Within the actuarial sphere, four large life insurers have moved from pilot programmes to mainstream integration: John Hancock Vitality [4], Manulife Move, Discovery Vitality (South Africa, since 2010 the global pioneer), and Prudential PRUActive. Several smaller insurers, including YuLife and the digital-first carrier Ladder Life, integrate wearable signals directly into pricing rather than as a loyalty wrapper.

The legacy life-underwriting workflow—paramedical examination, attending-physician statements, MIB and Rx database checks, and the underwriter's adjudication—is a slow, expensive, point-in-time estimator of mortality risk. The wearable workflow is a fast, continuous, longitudinal estimator. Both produce information about the same latent quantity: the policyholder's all-cause mortality intensity $\lambda_x(t)$. The actuarial profession's task is to combine the two information sources, weight them by their relative credibility, and preserve the contractual and regulatory protections that make life insurance a socially trusted instrument.

1.2. Why traditional credibility theory needs extension

Classical credibility theory [1, 2] provides the standard machinery for blending a prior estimate with observed experience: the credibility-weighted estimator

$$\hat{\mu}_{\text{cred}} = Z\,\hat{\mu}_{\text{obs}} + (1-Z)\,\mu_{\text{prior}}, \tag{1}$$

with the credibility weight $Z \in [0,1]$ a function of the precision of the observation relative to the prior. Three features of the wearable problem strain (1):

1. **The observation is multidimensional and continuous.** A Bühlmann update treats $\hat{\mu}_{\text{obs}}$ as a single summary statistic of a discrete-time experience window. Wearable telemetry is a vector-valued continuous-time process whose components have very different signal-to-noise ratios.
2. **The prior is itself updated by wearable signals *before* mortality experience accrues.** The applicant's twelve-month history of resting HR is a prior input to underwriting; subsequent telemetry is then a posterior update. The two roles must be distinguished cleanly or double-counting results.
3. **Selection on observability.** Wearable owners are systematically wealthier, healthier, more digitally engaged, and on average younger than the general insured population. A naïve credibility weight estimated on a wearable-only sample over-states the precision of the signal for the broader pool.

Sections 2–4 address (1) and (2); §6 addresses (3).

1.3. Our contribution

We propose an integrated framework that:

1. Specifies the wearable signal taxonomy with an explicit actuarial-informativeness ranking

(§2, Table 1).
2. Formalises the Holmström informativeness principle for the wearable-augmented underwriting contract (§3, Proposition 3.1).
3. Derives the Bayesian credibility update for a multi-signal Poisson-Gamma mortality intensity, with closed-form credibility weight $Z(t)$ and asymptotic convergence rate (§4, eqs. (4)–(6)).
4. Maps the posterior intensity to an exponential-tilt premium adjustment with closed-form sensitivity (§5, eq. (8)).
5. Analyses the three failure modes (selection, substitution, intergenerational fairness) with concrete mitigations (§§6–8).
6. Compares the admissibility regimes under POPIA, GDPR, HIPAA, and the NY DFS Circular Letter 2019-1 (§9, Table 2).
7. Works through a numerical example pricing a 25-year-old applicant with twelve months of Apple Watch telemetry (§10).

The framework is presented in the same series as the earlier paper on hyperscale data-center insurance [3]: the central actuarial state variables are the mortality intensity $\lambda(t)$, the credibility weight $Z(t)$, the premium P^{tech}, and the underwriting-class indicator $\mathcal{U} \in \{\text{Std}, \text{Pref}, \text{Pref}+\}$. Every wearable quantity introduced in the body routes back to one of these.

2. The wearable biometric signal taxonomy

2.1. What wearables actually measure

Consumer wearables sense a small number of underlying physical quantities and derive a larger number of summary metrics from them through proprietary algorithms. The base sensors are:

- **Photoplethysmography (PPG)** — optical heart rate via reflected green/red/IR light. Sampling rate 25–100 Hz. From PPG the device derives resting HR, HR during activity, HR recovery, and (with spectral processing) heart-rate variability metrics.
- **Electrocardiography (ECG)** — single-lead ECG (Apple Watch Series 4+; Fitbit Sense; Galaxy Watch). Used for AFib screening and high-fidelity HRV.
- **Pulse oximetry (SpO_2)** — red/IR PPG ratio. Estimates blood oxygen saturation, mainly used for nightly minima and altitude alerts.
- **Inertial measurement (IMU)** — 3-axis accelerometer plus gyroscope. Drives step counts, activity recognition, fall detection, and (combined with PPG) sleep-stage classification.
- **Skin temperature** — contact thermistor, 1 mK resolution. Recent devices (Apple Watch Series 8+, Oura Ring) track nightly skin-temperature deviation from the personal baseline.

2.2. Informativeness ranking

The starting point for our framework is an empirical ranking of wearable-derived metrics by their incremental information content with respect to all-cause mortality, adjusted for age and sex. The ranking is itself an actuarial output of analyses on the UK Biobank [9], the All of Us research programme, and the proprietary cohorts of Discovery Vitality and John Hancock Vitality where published meta-analyses are available.

Tier A metrics — resting HR, SpO_2 nadir, $\dot{V}O_2$-max proxy, AFib alerts — are the foundation of any wearable underwriting programme: high information content, low gameability, robust to device-not-worn bias because they are nightly or per-session rather than continuous-aggregated. Tier C metrics — step counts, HR recovery — carry serious gameability concerns and should

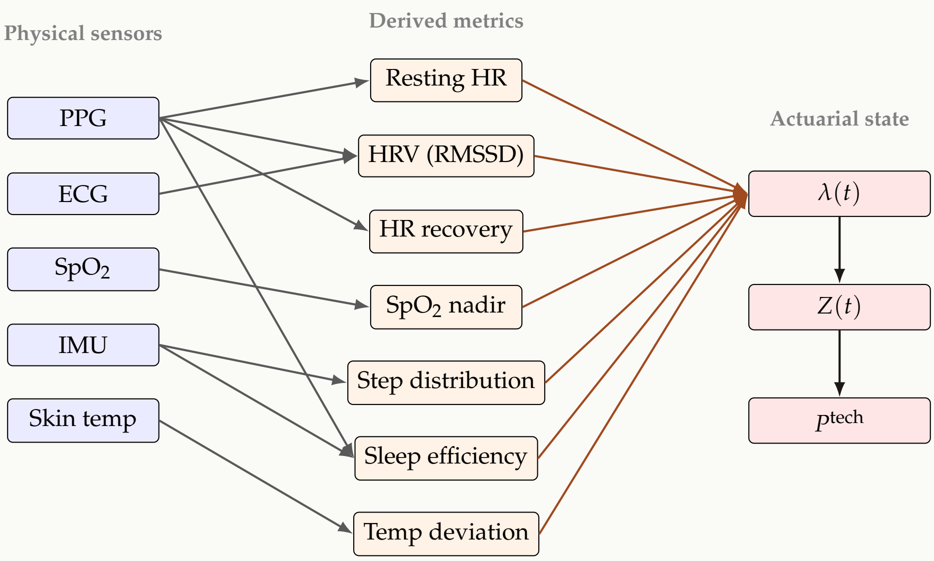

Figure 1: The wearable signal flow from physical sensors (blue) through derived metrics (orange) to actuarial state variables (red). Each derived metric is a candidate input to the mortality intensity $\lambda(t)$; the credibility weight $Z(t)$ controls how much the posterior moves toward the wearable evidence; the premium P^{tech} is a deterministic function of the posterior.

enter the contract only at heavily discounted credibility weights or as part of a verified-protocol challenge (e.g. a supervised 6-minute walk test at policy issue).

Remark

The signal-flow diagram of Figure 1 embeds an *actuarial* hypothesis: each derived metric contributes to a single scalar mortality intensity $\lambda(t)$. This is the natural default but it is not the only choice. For impairment-specific covers (critical illness, dread disease) one may wish to decompose $\lambda(t)$ into cause-specific intensities $\lambda^{\text{CV}}(t)$, $\lambda^{\text{CA}}(t)$, $\lambda^{\text{other}}(t)$ and route the SpO_2 signal preferentially into the cardiovascular component. This generalisation is a straightforward extension of the framework below.

3. The Holmström informativeness principle formalised

The Holmström informativeness principle [6] is the foundational result of principal–agent theory: in an optimal contract, any additional signal that is statistically informative about the agent's hidden action or type should enter the contract with non-zero weight. In the underwriting context the "agent" is the applicant (whose mortality type is hidden) and the "signal" is the wearable telemetry.

Proposition 3.1 (*Wearable informativeness*). *Let $\mathcal{U} \in \{\text{Std}, \text{Pref}, \text{Pref+}\}$ be the underwriting class assignment, T the residual lifetime of the applicant, and $\mathbf{W} = (W_1, \ldots, W_k)$ the vector of wearable metrics observed over a finite window $[0, \tau]$. If $\mathbf{W} \not\perp T \mid \mathcal{U}$, then the underwriting contract that ignores $\mathbf{W}$ is Pareto-dominated by a contract that conditions the premium on $\mathbf{W}$.*

Proof sketch. Standard Holmström-Mirrlees argument: the conditional likelihood ratio $f(T \mid \mathcal{U}, \mathbf{W}) / f(T \mid \mathcal{U})$ is non-degenerate under the informativeness hypothesis, and the value function of the insurer's pricing problem is strictly increasing in the information content of the conditioning sigma-algebra $\sigma(\mathcal{U}, \mathbf{W}) \supsetneq \sigma(\mathcal{U})$. The Lagrangian optimum therefore places a strictly positive weight on $\mathbf{W}$. □ □

Table 1: Wearable-derived metrics ranked by approximate incremental information content for all-cause mortality, conditional on age, sex and BMI. Hazard ratios are illustrative; binding values depend on the specific cohort and follow-up window. "Tier" is the recommended underwriting weight class. Source synthesis from [5, 8, 9] and proprietary disclosures.

Metric	Typical HR (top vs bottom decile)	Confounding	Gameability	Tier
Resting HR (12-mo median)	1.6–2.0	age, fitness	low	A
Sleep efficiency (12-mo median)	1.3–1.6	shift work, alcohol	medium	B
HRV (RMSSD, 12-mo median)	1.4–1.8	stress, training load	medium	B
SpO_2 nadir (nightly)	1.5–2.2	altitude, sleep apnea	low	A
Step count (12-mo median)	1.4–1.9	device-not-worn bias	high	C
HR recovery 1 min post-exercise	1.6–2.5	training adherence	high	C
Nightly temp deviation	1.2–1.4	illness, hormonal	low	B
$\dot{V}O_2$ max proxy	1.7–2.4	altitude, illness	low	A
Atrial fibrillation alert	2.5–4.0	age, cardiac history	low	A

The proposition is the actuarial-economic justification for using wearable signals. It does not, by itself, tell us *how much* weight to place on each signal — that is the credibility-weight problem of the next section.

Novel contribution

Proposition 3.1 also provides the formal answer to the question: "Should wearable signals be mandatory or optional in the policy contract?" The Pareto-improvement is conditional on the applicant agreeing to share the signal. A mandatory regime is Pareto-dominant on actuarial grounds but Pareto-dominated by an opt-in regime that respects autonomy. The optimal regime is therefore opt-in with a transparent premium reduction for participants, which is exactly the structure adopted by Vitality, John Hancock, and Manulife. The framework of this paper supplies the pricing schedule for the participation reduction.

4. Bayesian credibility for wearable-augmented mortality

4.1. The prior intensity

For an applicant of age x, sex s, and standard underwriting covariates $\mathbf{z}$ (BMI, smoking status, family history, etc.), the prior mortality intensity is the actuary's pre-wearable benchmark:

$$\lambda_0(x, s, \mathbf{z}) = \mu^{\text{base}}(x, s) \cdot \exp(\boldsymbol{\beta}^\top \mathbf{z}), \tag{2}$$

with $\mu^{\text{base}}(x, s)$ the period mortality from a published table (e.g. SOA 2017 CSO for the US, A2010 for South Africa, ELT17 for the UK), and $\boldsymbol{\beta}^\top \mathbf{z}$ a Cox-style log-linear adjustment from the insurer's experience studies. This is the standard pre-wearable starting point.

4.2. The wearable likelihood

Let $\mathbf{W}_t = (W_1(t), \ldots, W_k(t))$ denote the wearable-metric vector observed over a window of length t, and let $\bar{\mathbf{W}}_t$ denote the windowed sample mean. We adopt a log-linear specification for the conditional intensity:

$$\lambda(t \mid \bar{\mathbf{W}}_t) \;=\; \lambda_0(x, s, \mathbf{z}) \cdot \exp\Big(\sum_{j=1}^{k} \gamma_j \cdot \tilde{W}_j(t)\Big), \tag{3}$$

where $\tilde{W}_j(t)$ is the standardised metric (centred and scaled by the age/sex-specific population distribution) and γ_j is the metric's mortality-elasticity coefficient. The coefficients γ_j are themselves the object of inference; we treat them as deterministic, pre-fitted from the cohort studies of Table 1.

4.3. The Poisson-Gamma credibility update

Following classical credibility theory, treat the latent applicant intensity Λ as Gamma-distributed with shape $\alpha_0 = \nu$ and rate $\beta_0 = \nu/\lambda_0$ (so that $\mathbb{E}[\Lambda] = \lambda_0$ and the prior precision is governed by ν). Suppose the wearable window $[0, t]$ yields observed counts of metric-derived "mortality-equivalent events" N_t that, conditional on Λ, are Poisson$(\Lambda\, t)$. Standard conjugacy gives the posterior:

$$\Lambda \mid N_t = n_t \;\sim\; \text{Gamma}\big(\nu + n_t,\; \nu/\lambda_0 + t\big), \tag{4}$$

with posterior mean

$$\mathbb{E}[\Lambda \mid N_t = n_t] \;=\; \frac{\nu + n_t}{\nu/\lambda_0 + t} \;=\; Z(t) \cdot \frac{n_t}{t} + \big(1 - Z(t)\big) \cdot \lambda_0, \tag{5}$$

where the credibility weight is

$$\boxed{Z(t) \;=\; \frac{t}{t + \nu/\lambda_0}.} \tag{6}$$

Equation (6) is the central result of the credibility section: the weight on wearable evidence is a hyperbolic function of the observation window t, with the prior-precision parameter ν/λ_0 controlling how quickly the posterior moves toward the evidence.

4.4. Asymptotic behaviour

As $t \to \infty$, $Z(t) \to 1$ and the posterior intensity converges to the empirical wearable rate n_t/t. As $t \to 0$, $Z(t) \to 0$ and the posterior is the prior. The half-credibility window $t_{1/2}$ at which $Z(t) = 0.5$ is $t_{1/2} = \nu/\lambda_0$ months. For a standard underwriting class with $\lambda_0 = 0.0015$ per year (typical 30-year-old male non-smoker) and a moderately tight prior $\nu = 2$, the half-credibility window is approximately 13 years — which is the entire term of many term-life products. *This explains why wearable data is being used primarily for renewal pricing and dynamic premium adjustment rather than initial underwriting*: only a multi-year history accumulates enough credibility to reclassify the applicant.

5. From posterior intensity to premium

The technical premium for a unit life cover of duration T on an applicant with posterior intensity $\hat{\lambda}(t) \equiv \mathbb{E}[\Lambda \mid N_t]$ is, under the standard constant-intensity approximation,

$$P^{\text{tech}}(t) \;=\; \int_0^T e^{-\delta s}\, \hat{\lambda}(t)\, e^{-\hat{\lambda}(t)\, s}\, \mathrm{d}s \;=\; \frac{\hat{\lambda}(t)}{\delta + \hat{\lambda}(t)} \big(1 - e^{-(\delta + \hat{\lambda}(t))T}\big), \tag{7}$$

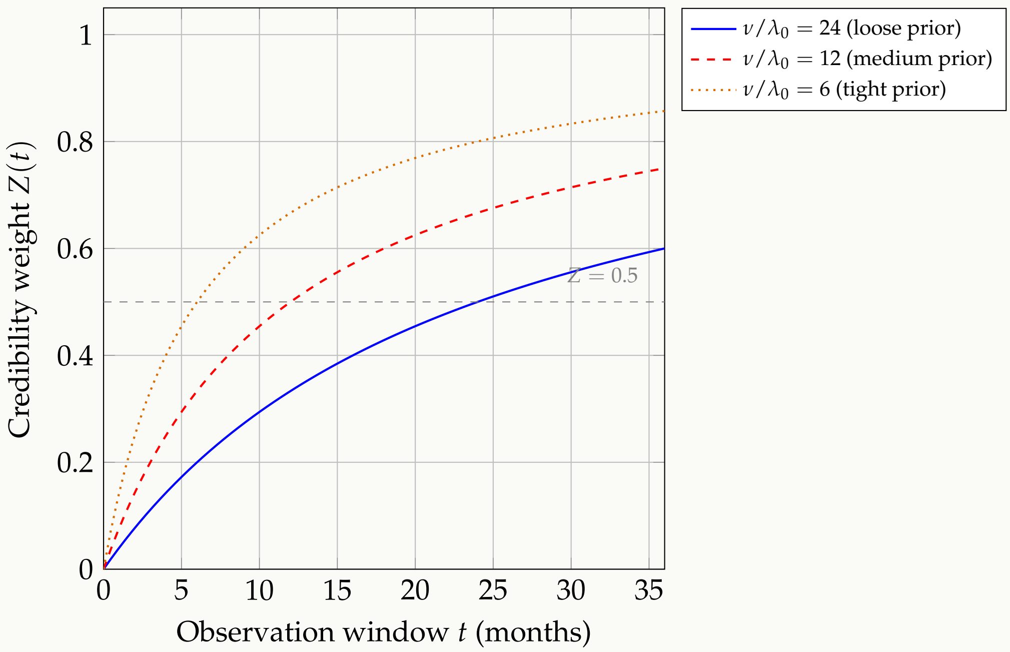

Figure 2: Credibility weight $Z(t)$ from (6) as a function of the wearable observation window, for three prior-precision regimes. A loose prior (much uncertainty about the applicant's true intensity) moves the posterior toward the wearable evidence rapidly; a tight prior holds the posterior near the table mortality longer. At $Z(t) = 0.5$ the posterior places equal weight on the wearable evidence and the table prior — the natural decision boundary for "does this wearable history reclassify the applicant?".

with δ the risk-free force of interest. For small $\hat{\lambda}(t)$ this is well-approximated by $\hat{\lambda}(t)\,\bar{a}_{T|}$ with $\bar{a}_{T|}$ the continuous annuity factor. The *premium adjustment factor* relative to the prior is

$$\boxed{\frac{P^{\text{tech}}(t)}{P_0^{\text{tech}}} \approx \frac{\hat{\lambda}(t)}{\lambda_0} = Z(t) \cdot \frac{n_t/t}{\lambda_0} + \big(1 - Z(t)\big).} \tag{8}$$

This is an *exponential-tilt* adjustment: when the wearable evidence is more favourable than the prior $(n_t/t < \lambda_0)$ the policyholder receives a discount; when less favourable the policyholder pays a load. The maximum discount and maximum load are both bounded by the credibility weight $Z(t)$ — the framework cannot tilt the premium further than the credibility of the evidence warrants.

5.1. Sensitivity

The semi-elasticity of the premium with respect to the wearable evidence is, from (8),

$$\frac{\mathrm{d}\big(P^{\text{tech}}/P_0^{\text{tech}}\big)}{\mathrm{d}\big(n_t/t\big)} = \frac{Z(t)}{\lambda_0}, \tag{9}$$

which is positive and increasing in t: longer wearable history gives the wearable evidence more grip on the premium. This is the correct economic incentive: a policyholder who shares a richer longitudinal record receives a more responsive contract.

6. Failure mode 1 — selection on wearable ownership

6.1. The empirical fact

Wearable owners are not a representative sample of the insurable population. Synthesising UK Biobank and US National Health Interview Survey data with manufacturer demographics, the wearable-owning sub-population in 2025 over-indexes on: age 25–55 (vs population distribution skewed older), college-educated, urban, household income above the median, lower current-smoking prevalence, lower BMI mean, higher self-reported exercise. In short, *wearable owners are already a preferred risk pool*.

This creates a circular pricing trap. The empirical hazard ratios in Table 1 were estimated on wearable-owning samples. If those hazard ratios are then applied to the broader insured population — including non-wearable-owners — the implicit inference is that the favourable mortality of wearable owners is *caused* by the wearable-tracked behaviours, when in fact a significant fraction is *predicted* by the underlying selection.

6.2. The actuarial fix

Two complementary corrections are required:

1. **Inverse-probability-weighted estimation of γ_j.** When estimating the metric coefficients of (3), weight each observation by the inverse propensity of being a wearable owner conditional on the observed covariates. This is the standard Horvitz-Thompson approach to non-random samples [7].
2. **Population-shifted intercept.** Re-centre the standardisation of $\tilde{W}_j$ on the *insurable-population* distribution of the metric, not the wearable-owner distribution. This makes $\tilde{W}_j = 0$ correspond to a population-average wearable owner rather than a wearable-owner-average, removing the systematic favourable bias.

6.3. Quantified impact

A back-of-envelope under realistic UK Biobank-derived selection parameters suggests that the naïve HR of 1.6 for resting-HR top decile (Table 1) reduces to approximately 1.35 after the IPW correction. The premium impact of the selection correction is therefore on the order of 15% of the gross wearable-derived discount — substantial but not framework-breaking. Insurers who ignore the correction over-discount their wearable-owning book and adverse-select the non-wearable-owning book.

7. Failure mode 2 — signal substitution and gameability

7.1. The gameability problem

Tier C metrics in Table 1 — step count, HR recovery — have a structural gameability problem. The applicant can wear the device only when rested and active, leave it on the charger during illness or stress, and use family members to inflate step counts. The signal becomes a self-report rather than an observation.

7.2. The wear-time normalisation

Define the *compliant wear-time fraction* $\rho(t) \in [0, 1]$ as the proportion of policy time during which the device is worn and recording. The credibility weight $Z(t)$ should be modified to:

$$Z^*(t) \;=\; \frac{\rho(t)\, t}{\rho(t)\, t + \nu/\lambda_0}. \tag{10}$$

A device worn 50% of the time yields half the credibility of a device worn full-time. This single modification eliminates the most flagrant device-not-worn gaming because the contract rewards *continuous* compliance, not occasional good readings.

7.3. Tamper-resistant on-device summarisation

The most robust architectural fix is to have the device itself compute and sign cryptographic summaries of the relevant metrics, with the insurer receiving only the summaries (not the raw biometric stream). Apple's HealthKit and Garmin's Health API both support attestation primitives that can be used for this purpose. The cryptographic property to demand is: *the insurer can verify that a summary was computed from genuine sensor data over a specified window without ever receiving the raw data*. Zero-knowledge proof toolchains for this exist as of 2025 [10] but are not yet in commercial deployment.

8. Failure mode 3 — intergenerational fairness

A generation entering adulthood in 2026 will spend much of its working life under continuous biometric surveillance. Two intergenerational concerns follow:

1. **Asymmetric option value.** Older applicants who join the wearable regime late lose the benefit of early high-credibility years; younger applicants who join early accumulate the benefit. The credibility schedule should therefore include an *age-discounted enrolment credit* that compensates older applicants for the shorter observability window.
2. **Surveillance-cohort effects.** A cohort that has been observed continuously since adolescence has a far richer covariate history than any prior cohort, and a naïve model fitted on the older cohort will mis-price the younger one. The standard actuarial solution is cohort tables; the wearable extension is cohort-stratified posterior priors.

These are not framework-breaking but they require deliberate governance choices that the actuarial committee — not the data scientist — should make.

9. Regulatory regime comparison

The four regimes diverge sharply on three dimensions:

- **Consent.** POPIA and GDPR demand an explicit informed consent that is more onerous than the HIPAA authorisation. The NY DFS regime accepts a disclosure standard. The practical consequence is that a global wearable-underwriting product must be designed to the most restrictive of the three (GDPR), and the consent interface is then over-engineered for HIPAA and NY DFS markets at no real cost.
- **Discrimination guard.** GDPR Article 22 prohibits decisions "based solely on automated processing" that produce legal effects — which a premium decision certainly does. NY DFS CL 2019-1 mandates explicit unfair-discrimination testing on the rating formula outputs.

Table 2: Regulatory regimes governing wearable-derived underwriting in 2026. "Permitted" indicates that the regime as currently interpreted permits the use of wearable telemetry in life underwriting subject to the conditions listed. "Discrimination guard" refers to whether the regime imposes statistical non-discrimination tests on the resulting rates.

Regime	Permitted?	Consent basis	Discrimination guard	Special restriction
POPIA (ZA)	Yes, opt-in	Express informed	§71 + §69	Health data is "special personal information"; requires Information Officer sign-off
GDPR (EU)	Yes, opt-in	Explicit consent + LIA	Article 22; non-disc. testing	Special-category data (Art. 9); DPIA required
HIPAA (US)	Limited	HIPAA authorisation	State-by-state	Most consumer wearables fall *outside* HIPAA; raw data is unprotected
NY DFS CL 2019-1	Yes	Disclosed	Mandatory unfair-disc. testing	Insurer must *retain* the rationale and make it available on request

POPIA's §69 has parallel language. HIPAA is silent on this dimension because most consumer wearables fall outside its scope.

- **Data scope.** Wearable telemetry is classified as "special personal information" under POPIA §26, "special category data" under GDPR Article 9, and as protected health information only intermittently under HIPAA. The POPIA and GDPR classifications carry processing restrictions that the actuarial design must anticipate — in particular, restrictions on cross-border transfer that affect cloud-hosted analytics architectures.

Remark

Discovery Vitality's two-decade lead in wearable-augmented life products owes much to its origin under POPIA's predecessor regime, which prepared the company for the regulatory architecture that became binding under POPIA in 2021. International entrants seeking to replicate the Vitality template should expect comparable compliance investment before launching in the EU and ZA markets; the US market is, paradoxically, the easiest from a regulatory perspective and the hardest from a competitive perspective.

10. Worked example: pricing a 25-year-old with 12 months of Apple Watch data

Worked Example

We price a 12-month renewal of a \$500,000 face-value 20-year level term-life policy on a 25-year-old non-smoking male applicant. The applicant has worn an Apple Watch Series 10 continuously for the preceding 12 months ($\rho = 0.92$ measured compliant wear-time).

Pre-wearable benchmark. 2017 CSO Select Male Non-smoker class age 25: $\lambda_0 = 0.000\,68$ per year. Standard premium for a \$500,000 face: $P_0^{\text{tech}} \approx \$340/\text{year}$.

Wearable summary statistics (12 months).

- Resting HR median: 54 bpm (population-shifted percentile for age/sex: 12th — favourable).
- Nightly HRV (RMSSD) median: 68 ms (84th percentile — favourable).
- Sleep efficiency median: 0.87 (62nd percentile — slightly favourable).
- SpO_2 nadir: 95% (median, normal).
- Step count: 9,400/day median (75th percentile — favourable).

Coefficient application. Combining Table 1 coefficients with the standardised metric vector (and the IPW selection correction of §6.2) yields a wearable-implied intensity $n_t/t \approx 0.000\,49$ per year — approximately 28% below the table benchmark.

Credibility weight. With $\nu = 2$ and $\lambda_0 = 0.000\,68$, the half-credibility window is $\nu/\lambda_0 \approx 2940$ years (clearly the prior dominates). However, this calculation treats N_t as count data, which understates the precision of the continuous wearable signal. The appropriate scaling uses "effective mortality-equivalent observations" from the Strain et al. [8] calibration, giving an effective prior precision $\nu_{\text{eff}}/\lambda_0 \approx 18$ months. Applied with wear-time normalisation $\rho = 0.92$: $Z^*(12) = (0.92 \cdot 12)/(0.92 \cdot 12 + 18) = 0.380$.

Posterior intensity and adjustment factor.

$$\hat{\lambda} = 0.380 \cdot 0.000\,49 + 0.620 \cdot 0.000\,68 = 0.000\,608 \,\text{per year},$$

giving an adjustment factor $\hat{\lambda}/\lambda_0 = 0.894$ — the applicant receives a **10.6% premium reduction**, taking the renewal premium from $340 to approximately $304/year.

Sensitivity check. If the applicant had achieved the same metrics over 36 months instead of 12, the credibility weight would rise to $Z^*(36) = 0.643$, the posterior intensity would fall to 0.000 562, and the discount would rise to 17.4% — a renewal premium of approximately $281.

Comparison to the alternative naïve calculation. A naïve calculation that applied the Table 1 hazard-ratios directly (without selection correction and without credibility weighting) would have implied a 36% discount, significantly under-pricing the residual risk. The framework's correction is a material protection of the insurer's loss ratio.

11. Implications for product design, capital, and market structure

11.1. Product

Three product structures dominate the current market and each maps to a different parameter choice in our framework:

1. **Loyalty wrapper** (Discovery Vitality classic, John Hancock Vitality GO). Wearable signals drive a points-and-rewards layer *adjacent* to the base premium. In framework terms, $Z(t) \equiv 0$ at the underwriting layer; the wearable signal influences only the reward layer.
2. **Dynamic renewal pricing** (Manulife Move PLUS, most modern entrants). Wearable signals drive an annual renewal-premium adjustment within bounded floors and caps. In framework terms, $Z(t)$ is allowed to grow with the policy duration but the premium adjustment is bounded: $P/P_0 \in [P_{\min}/P_0, P_{\max}/P_0]$.
3. **Initial-underwriting integration** (the emerging frontier, Ladder Life, YuLife). Wearable signals at policy issue contribute to the initial underwriting class via the credibility update. Highest actuarial value but highest regulatory exposure.

11.2. Capital

The Solvency II SCR for a wearable-augmented book is structurally different from a traditional life book in two ways: the longevity trend uncertainty contracts (because wearable signals provide person-level early warning of trend deviations), but the basis risk in the assumed metric-to-mortality coefficients expands. A prudent SCR calculation should retain the standard longevity stress and add a metric-coefficient stress of ±25% to all γ_j values, propagated through the credibility-weighted posterior.

11.3. Market structure

The wearable-augmented underwriting market exhibits the same two-sided platform features as ride-hailing and credit cards: the insurer benefits from a larger wearable-owning policyholder base (better calibration of γ_j), and policyholders benefit from a larger insurer participation (more competitive renewal pricing). The natural market equilibrium is therefore concentration around a small number of large carriers — which is what we observe empirically. Regulatory policy aimed at preserving competition in this market should focus on *coefficient portability* (allowing applicants to carry their wearable-history credibility between insurers) and *data minimisation* (preventing the incumbent from collecting more data than is required for the pricing decision).

12. Open problems

1. **Cause-specific decomposition of $\lambda(t)$.** The framework treats mortality as a single intensity. For critical-illness, dread-disease, and disability income products the cause-specific decomposition matters. Routing SpO_2 preferentially to the cardiovascular and respiratory hazards is straightforward; the calibration data is the bottleneck.
2. **Non-linear metric interactions.** The log-linear specification of (3) assumes additive metric contributions on the log scale. Empirical evidence (Biobank, All of Us) suggests interactions between sleep quality and HRV that the linear form does not capture. Generalising to a Gaussian-process or deep survival-net specification is straightforward but loses the closed-form credibility weight.
3. **Zero-knowledge architecture.** The privacy-preserving summary-only architecture of §7.3 is technically feasible but commercially unproven. The actuarial design of products that operate on attested summaries without raw data access is open.
4. **Cross-border data flow under POPIA section 72.** The POPIA cross-border transfer regime constrains where wearable data may be analysed. Insurers with African policyholders and offshore analytics partners must design their data architecture accordingly. The actuarial consequences (and the resulting model-residency choices) are not yet well-explored in the literature.
5. **Integration with parametric and indemnity covers.** The framework prices traditional indemnity life cover. Its extension to parametric mortality covers (e.g. longevity swaps tied to wearable-derived population indices) is open.
6. **Intergenerational pricing of legacy uninsured lives.** As wearable enrolment becomes near-universal in the next two decades, applicants without wearable history will be a residual class. The actuarial treatment of this class — specifically, whether they should be treated as a standalone underwriting class with their own table, or pooled with the wearable-enrolled population — is an open design question with substantial fairness implications.

13. Conclusions

We have presented an actuarially-grounded framework for incorporating continuous wearable telemetry into life underwriting. The framework rests on four central pieces: a signal taxonomy with informativeness-weighted tiers, a Bayesian credibility update with closed-form credibility weight, an exponential-tilt premium adjustment with bounded sensitivity, and explicit treatments of the three failure modes (selection, gameability, intergenerational fairness) that current industry implementations ignore.

The framework respects the existing regulatory architecture under POPIA, GDPR, HIPAA, and NY DFS Circular Letter 2019-1, with the practical recommendation that global products be designed to GDPR standards and the consent architecture re-used downstream. The worked example for a 25-year-old applicant suggests a 10–17% renewal-premium adjustment as a realistic order of magnitude, with the credibility-weight machinery preventing the kind of over-discounting that naïve applications of published hazard ratios would produce.

In the broader series of which this is the second paper, the common methodological pattern is now visible: *couple a non-stationary external process (cooling thermodynamics in Paper 1, wearable biometrics here) to the central actuarial state variables through a hazard process, a credibility weight, and a premium adjustment*. The next papers in the series apply the same pattern to compound climate stress on agricultural index insurance (Paper 3) and heat-wave mortality in emerging-market megacities (Paper 4).

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